

Off-Ball Movement Direction and Localized Defensive Reorganization in Football

Introduction

A defence can shift together while nearby defenders also change position within the unit. We call movement by nearby defenders relative to the wider defensive unit localized defensive reorganization. Distinguishing these geometries lets us ask whether off-ball movement away from the pitch centreline (“outward”) is associated with different subsequent reorganization than comparable movement toward goal.

Methods

Fixed, non-overlapping two-second intervals measured attacker movement and subsequent defender-relative path. We compared the average accumulated defender-relative path of the nearest three defenders with that of four middle-ranked defenders, fixing proximity ranks before the defender interval without inferring marking assignments. The temporal design was developed in Metrica Game 1, tested in held-out Game 2, and evaluated externally in seven IDSSE matches using a paired reverse-time comparison. Directional models conditioned on attacker path magnitude and starting geometry, with a separate nine-match SkillCorner replication and a prospectively frozen replication in ten additional nonoverlapping SkillCorner matches; cohorts and tracking environments were analysed separately, not pooled. Uncertainty was estimated by block bootstrap, preserving simultaneous attacker perspectives and temporal grouping rather than treating individual observations as independent.

Results

Outward movement was associated with greater subsequent localized reorganization than goalward movement in all three separately analysed cohorts. For the prespecified comparison of straight 5 m outward versus straight 5 m goalward movement, holding modeled path magnitude and context equal, the near-minus-middle path difference was approximately 28 cm in IDSSE (95% interval: 26–31 cm) and 24 cm in SkillCorner (21–27 cm). These are differences between group-average defender-relative paths, not additional movement by each defender. Directional contrasts were positive in all seven IDSSE and all nine original SkillCorner matches. In the prospectively frozen additional SkillCorner population, the contrast was 0.05501 m/m [0.05006, 0.06002], with positive match and leave-one-match-out estimates in all 10/10 matches. The temporal near-minus-middle association also replicated from Metrica to IDSSE (0.06115 m/m, 95% CI [0.05579, 0.06681]) and was positive in all seven IDSSE matches. The IDSSE forward-minus-reverse excess was 0.02455 m/m [0.01932, 0.02985], positive in all seven matches, although reverse-time structure remained positive. A secondary analysis associated goalward movement more strongly with collective defensive translation; the proposed width-narrowing explanation was not established.

Conclusion

The findings show that goalward progression and localized defensive reorganization are distinct geometric descriptions of off-ball movement. They do not establish causation, marking responsibility, tactical effectiveness or attacking value. The measurement could provide a descriptive layer for organizing candidate passages for subsequent video review; its football interpretation and review usefulness remain unvalidated.

Time-ordered localized defensive reorganization

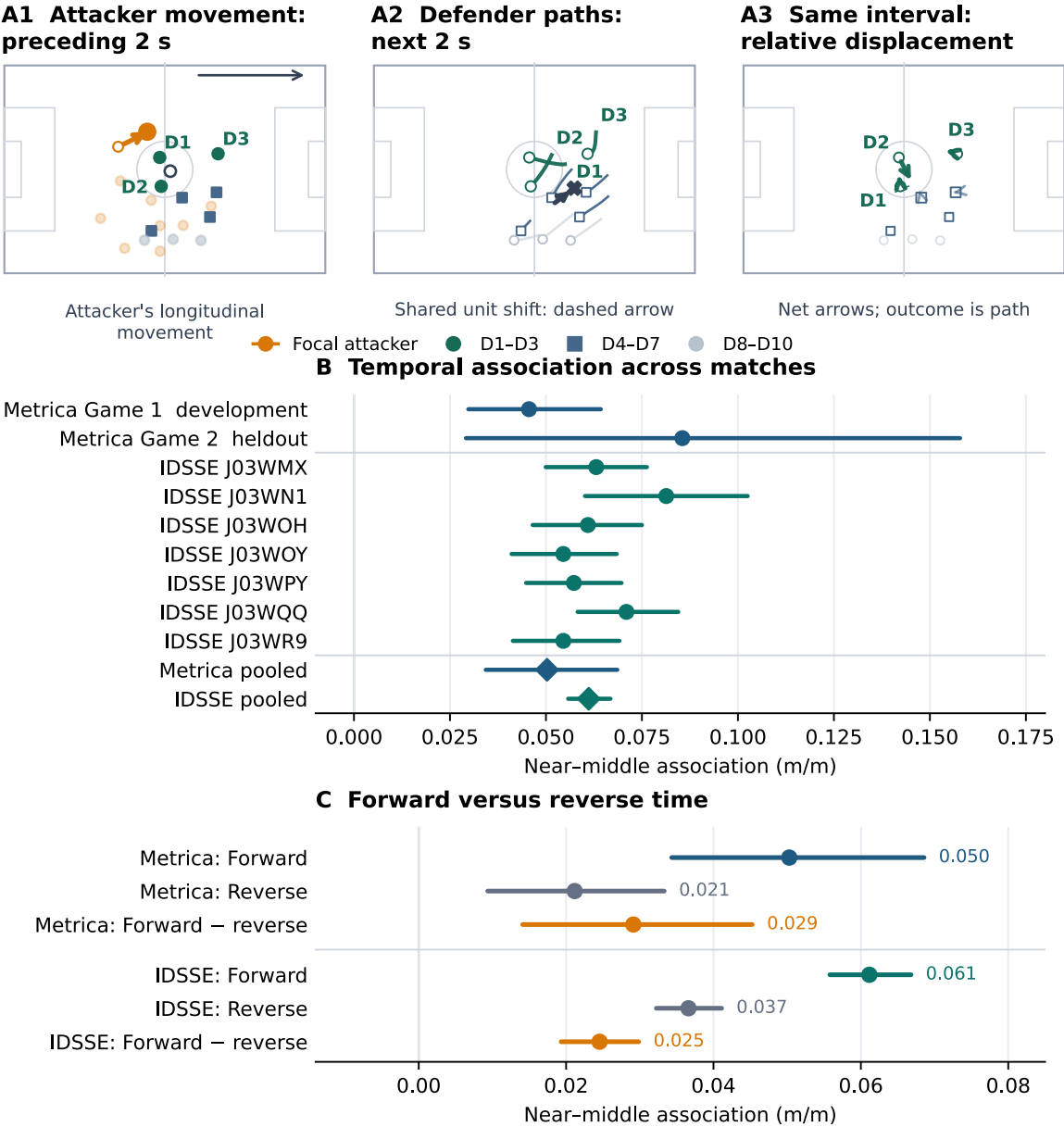


Figure 1. Time-ordered localized defensive reorganization. A1 shows preceding attacker movement; A2 shows absolute defender paths over the next two seconds; A3 shows net defender displacement relative to the other nine defenders over that same interval. The statistical outcome is accumulated defender-relative path, rather than the illustrated net arrows. The Metrica Game 2 example (period 1, anchor 2336.04 s) was selected as the earliest eligible anchor at or above the upper quartile of preceding attacker path, using attacker movement only. B shows match estimates and separate within-environment pooled estimates. Near-middle compares D1–D3 with D4–D7; m/m denotes metres of defender-relative path per metre of preceding attacker path. C shows positive forward-minus-reverse excess despite positive reverse-time structure. Intervals are 97.5% for Metrica and 95% for IDSSE. The example and associations establish neither marking assignments nor causation.

Directional difference replicates across tracking environments and additional matches

Outward: away from the pitch centreline · Goalward: toward goal

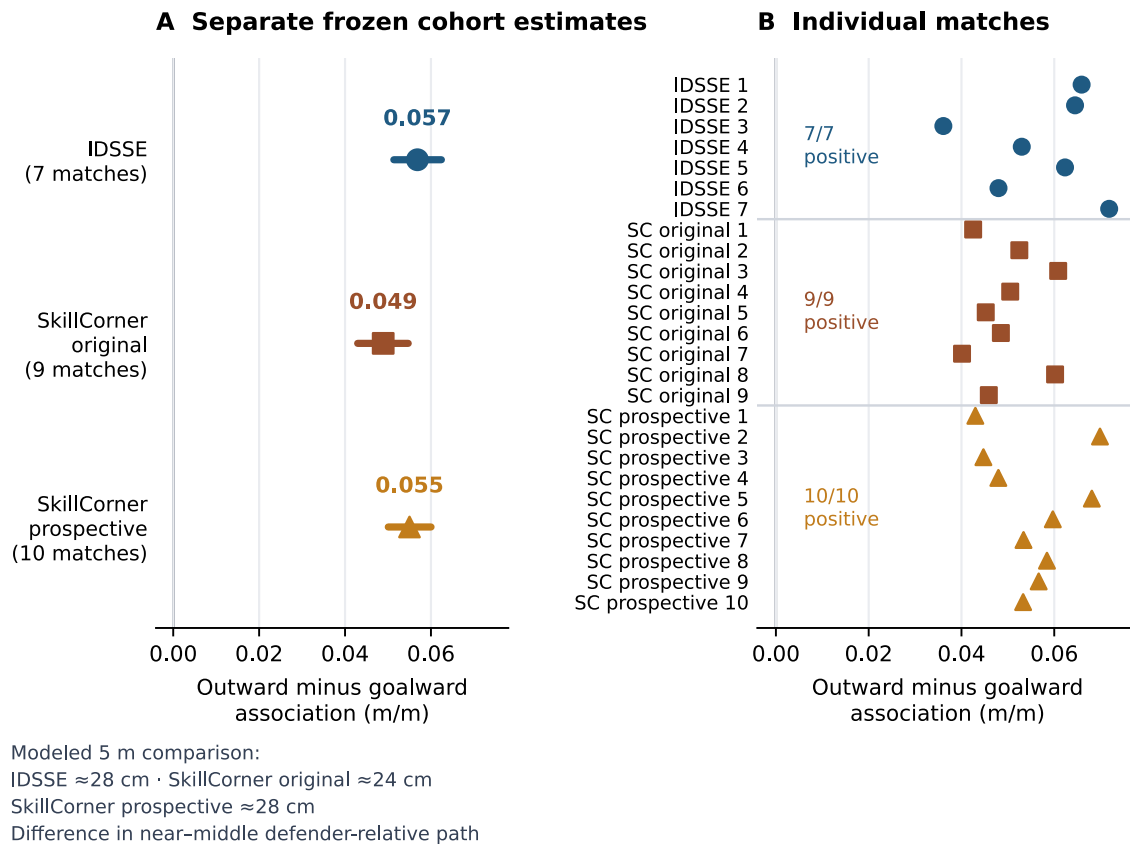


Figure 2. Replicated outward-versus-goalward difference in localized defensive reorganization. Outward means away from the pitch centreline; goalward means toward goal. A shows separately estimated conditional contrasts: IDSSE 0.056856 m/m (95% CI [0.051358, 0.062430]), original SkillCorner 0.048883 m/m [0.042940, 0.054707], and the prospective additional SkillCorner cohort 0.055007 m/m [0.050064, 0.060021]. For straight 5 m outward versus straight 5 m goalward movement, holding modeled path magnitude and starting context equal, these correspond to approximately 28 cm, 24 cm, and 28 cm greater near-minus-middle group-average defender-relative path. They do not represent extra movement by each defender. B shows all 7/7 IDSSE, 9/9 original SkillCorner, and 10/10 prospective SkillCorner match contrasts positive. The cohorts and providers were not pooled. The results describe observational geometry, not causal influence, tactical effectiveness or value.